

Learning by Teaching: Designing Teachable Agents to Support Children's Pronunciation Skill Learning

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Abstract

Pronunciation is a critical yet challenging aspect of language acquisition, especially for children. In this work, we propose a new teachable agent system designed to support children's pronunciation learning through a Learning-by-Teaching paradigm to address this, featuring the Phoneme-level Mispronunciation Projection (PMP) method. This method enhances learning by having the agent reproduce or exaggerate children's mispronunciations, encouraging their correction and practice. We implemented a prototype system featuring both virtual and physical embodiments of teachable agents, allowing us to explore their effectiveness in motivating children. In a pilot study with 27 children aged 7–9, we evaluated the PMP method's perceived performance, children's motivation to teach, and the influence of agent embodiment (virtual vs. physical robots). Results demonstrate the effectiveness of PMP in supporting pronunciation learning and highlight the physical robot's advantages in fostering engagement. This work also offers initial design implications for future teachable agents to support children's pronunciation skill learning.

CCS Concepts

• **Human-centered computing** → **User studies**; *Sound-based input / output*; • **Social and professional topics** → **Children**; **K-12 education**.

Keywords

Learning by teaching, pronunciation learning, Child-Computer Interaction, social robots

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1 Introduction

Pronunciation is a fundamental aspect of language acquisition, yet it is frequently overlooked in educational settings [5, 9]. This oversight is particularly concerning for young learners, who often struggle to produce sounds and words accurately in the target language. In the early stages of learning, young learners face two interconnected challenges: their inability to effectively correct pronunciation errors often leads to communication breakdowns and a growing sense of frustration [1]. This frustration, compounded by perceived lack of progress, can significantly demotivate learners, causing them to avoid practicing pronunciation altogether. This avoidance perpetuates a vicious cycle that hinders their linguistic development and overall confidence in language use.

Restoring the motivation and confidence to practice pronunciation is crucial to breaking this cycle [27]. Learning by teaching (LBT) [7], a well-established pedagogical method where the student acts as a teacher to instruct others has shown significant promise in addressing such challenge. By engaging in the teaching process, children can develop a deeper understanding of target concepts, receive feedback, and gain a sense of ownership over their learning - all of which can boost their confidence and motivation to continue practicing.

To facilitate learning by teaching, researchers have developed Teachable Agents — computer systems designed to enable students to teach. These agents have been extensively studied and applied in various educational contexts [4, 18, 19], such as learning mathematics [13] and science [2]. However, their potential for pronunciation learning remains relatively unexplored.

Tanaka *et al.* utilized a humanoid robot as a pupil, asking children to teach it verbs [25]. While effective, the robot demonstrated only the actions associated with the verbs and did not engage in speech. Advancements in speech analysis and generation now enable teachable agents to analyze and produce speech, opening new possibilities for interactive and effective pronunciation learning.

In the computer-supported learning-by-teaching paradigm, agents can take either a virtual form, such as animated characters, or a physical form, such as robots. For instance, Biswas *et al.* [2, 12] developed Betty’s Brain, a teachable computer agent in the domain of river ecosystems, allowing students to construct causal concept models by teaching the agent. Similarly, Gargot *et al.* [8] proposed a teachable robotic agent for treating severe dysgraphia, enabling children to practice handwriting by teaching the robot.

Despite these advancements, the relative effectiveness of different teachable agent embodiments, such as virtual agents or physical robots, remains a complex topic that has been seldom explored in depth [11]. Specifically, it is still unclear which type of teachable agent is better suited to motivate young learners to practice pronunciation and achieve improved learning outcomes. This work addresses these gaps by preliminarily designing, implementing and evaluating a teachable agent prototype to help children learn pronunciation.

As a first step to designing an effective teachable agent for pronunciation learning of children, in light of the aforementioned challenges, we initially propose the following two key design considerations: 1) To enable young learners’ ability to correct their pronunciation errors, the learning-by-teaching activity should be tailored to their specific pronunciation difficulties and provide timely feedback to guide their teaching [3]; 2) To maintain young learners’ motivation and engagement, the design should leverage the *Protege Effect* [4], where the learner feels responsible for the agent’s learning and takes the teaching role as a way to improve their own skills - this can further strengthen their self-efficacy and confidence.

In this work, we aim to explore the preliminary design of a teachable agent system to support children’s pronunciation learning and establish design principles for future development. Specifically, we propose a Phoneme-level Mispronunciation Projection (PMP) approach to tailor the learning-by-teaching activity to children’s unique pronunciation difficulties. The PMP approach involves the teachable agent intentionally reproducing or amplifying the same mispronunciations as the children. Following this, the children are tasked with teaching the agent the correct pronunciation, reinforcing their own learning through the process. Based on this approach, we pre-designed and implemented a prototype system including one virtual form and one physical robot form of the teachable agent, which serves as the starting point for the planned iterative design study. Through a pilot study with 27 children, we evaluate our prototype to address the following research questions:

RQ1: How do children perceive the accuracy of the implemented PMP method in the learning-by-teaching activity?

RQ2: Are children motivated to teach the agent pronunciation in the current design?

RQ3: What form of teachable agent - virtual or physical robot - can more effectively motivate children in the learning-by-teaching activity?

RQ4: What design considerations are crucial for developing effective teachable agents for children’s pronunciation learning?

The contribution of this work is two-folds: 1) A preliminary design of a teachable agent system to support children’s pronunciation learning, including the proposed Phoneme-level Mispronunciation Projection approach; 2) Pilot study results that provide insights on the efficacy of the design and reveal considerations for future development of teachable agents for pronunciation learning.

2 Interaction Design

2.1 Phoneme-level Mispronunciation Projection

Difficulties in second-language (L2) pronunciation often arise from perceptual challenges, as learners struggle to distinguish phonetic differences between their native language (L1) and L2 [24]. Perceptual training focused on L2 sounds has been shown to significantly enhance production abilities [23]. Techniques such as exaggerated audio pronunciations further aid learners by amplifying specific phonetic elements, enabling better error detection and correction [3, 6]. Building on this foundation, our design introduces Phoneme-level Mispronunciation Projection (PMP), where the system amplifies the teachable agent’s pronunciation errors to reflect the user’s mispronunciations. This personalized approach helps users recognize and correct their mistakes by teaching the agent the correct pronunciation. The PMP method consists of two phases. First, the system analyzes the user’s pronunciation at the phoneme level, identifying performance accuracy and *sound-like phonemes*—phonetically similar but incorrect pronunciations (e.g., mispronouncing *e* as *æ*). Second, the system selects words containing the target phoneme and generates augmented examples by replacing them with the corresponding *sound-like phoneme*. These examples help users perceive differences and practice correcting their pronunciation effectively.

2.2 Agents

Prior research on the LBT method suggests that student tutors learn more effectively when observing their tutees applying the knowledge they have taught [20]. This process benefits from a third-party mentor character to evaluate and guide the teaching. To this end, our design incorporates two intelligent agents: a mentor agent and a teachable agent. Embodied agents have also been shown to enhance engagement and performance in interactive tasks like tutoring [11, 25]. One aim of this study is to compare different embodiments of the teachable agent:

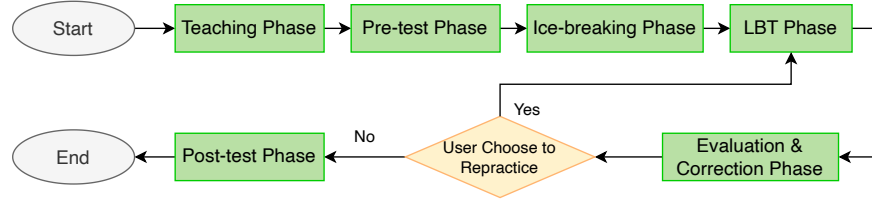


Figure 1: User Interaction Flow Design

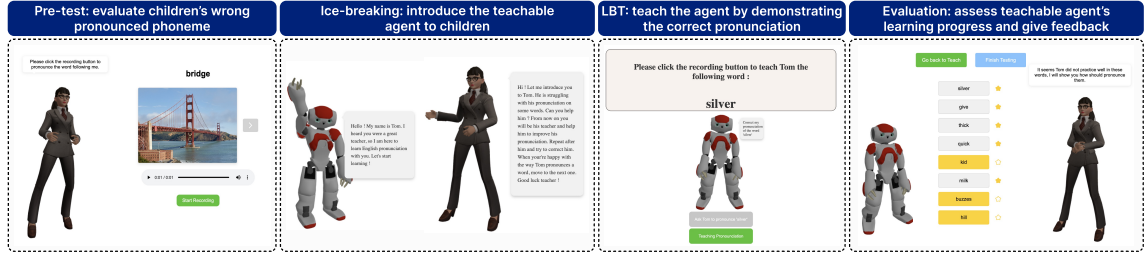


Figure 2: Core LBT Interactive Phase Design

a 3D virtual avatar and a physical robot. A humanoid robot was chosen for the physical embodiment due to its ability to play synthesized audio and perform gestures, adding realism and engagement. For example, during the LBT phase, the teachable agent may express confusion to prompt user assistance or display celebratory gestures during the evaluation phase, fostering engagement and a sense of accomplishment. These physical behaviours are also replicable in the virtual agent. The mentor agent, represented as a 3D virtual character with simple animations, is designed to resemble an adult and act as a guiding figure while maintaining an approachable and supportive demeanour.

2.3 Interaction Design for Learning by Teaching Activity

The interaction flow consists of six phases, as Figure 1. Before the interaction begins, teachers set the target phonemes they want students to practice. This selection generates the list of words, their pronunciation recording, and the corresponding images used in the subsequent phases.

Teaching Phase: In this phase, the user listens to the pronunciation of selected words while viewing an image that illustrates each word. This combination of auditory and visual input helps simulate a natural learning process, making it easier for children to associate sounds with meanings. This approach has been shown to effectively enhance vocabulary acquisition and engagement [15].

Pre-test Phase: This phase assesses the user's pronunciation performance to identify two key aspects: (1) phonemes that require further practice and (2) common phoneme-mistake pairs for personalized correction. The user is presented with n_w words, including their recordings and associated illustrations, and is asked to repeat them. Among the target phonemes, the n_{ph}

lowest-performing ones are selected for practice. If a user's score for a targeted phoneme exceeds a predefined threshold, the system shifts focus to the next most problematic phoneme. Additionally, for each selected phoneme to work on, the system identifies n_{mp} associated mispronunciations for personalized training (see Section 2.1).

Ice-breaking Phase: This phase involves both the mentor and the teachable agent, aiming to foster a connection between the user and the latter. The mentor introduces the teachable agent, explaining its difficulties with certain pronunciations and encouraging the user to help it practice. The teachable agent engages the user with simple interactive behaviours, such as greeting gestures and a small prompt. It expresses gratitude, compliments the user as a good teacher, and seeks their assistance with pronunciation practice.

LBT Phase: This is the core phase of the interaction design. To ensure a focused teacher-student relationship and minimize distractions, the mentor agent exits, leaving the user to interact solely with the teachable agent to teach pronunciation. A key feature of this phase is the use of PMP, where the student agent's pronunciation errors are deliberately exaggerated (see section 2.1). Users can teach each word 3–6 times. For each new word, the teachable agent begins with an intentionally incorrect pronunciation. After each teaching attempt, the system evaluates pronunciation accuracy at the phoneme level. If the user achieves a phoneme accuracy score above a set threshold for the target sound in at least two attempts continuously, the teachable agent progresses to pronouncing the word correctly.

Evaluation and Correction Phase: In this phase, the mentor agent evaluates the teachable agent's progress and provides recursive feedback to the user [20]. The teachable agent's pronunciation reflects the final versions of words learned during the LBT phase, highlighting any remaining errors and encouraging the user to

revisit specific words. Words are pronounced one by one with on-screen animations indicating correctness, and a summary screen provides a binary status (correct/incorrect) for all words. Upon request, the mentor agent can demonstrate correct pronunciations to enhance learning. Users can choose to reteach words or conclude the session, with the design assessing their willingness to continue teaching based on the teachable agent's progress.

Post-test Phase: This phase evaluates the user's learning and practice outcomes during the study. This phase mirrors the pre-test phase, directly comparing pronunciation accuracy scores before and after the teaching interaction. By maintaining identical conditions, this comparison provides a clear assessment of the progress of the user's word and phoneme-level pronunciation.

2.4 Implementation

The implementation consists of four main components: the backend, interaction interface, digital avatars (mentor and virtual teachable agents), and the robot, all communicating via Flask-based APIs. The Python backend handles pronunciation analysis and audio generation, leveraging Azure's Pronunciation Assessment API [16] for phoneme-level metrics like word accuracy, phoneme accuracy, and "sounds-like" phonemes. Using these metrics, it identifies poorly performed phonemes and generates PMP audios through Azure's IPA Pronunciation API [17]. Fine-grained control is achieved via Speech Synthesis Markup Language (SSML), enabling precise phoneme specification using the International Phonetic Alphabet (IPA). Both virtual and physical agents use these audio files during interactions. The web-based interface, built with React, guides users through the interaction phases. It embedded virtual agents (mentor and teachable when applicable), represented as 3D-animated GLTF models. It also provides tools and UI for teaching, real-time feedback, and progress tracking. The teachable agent's physical embodiment is the humanoid robot Nao [21], which enhances engagement and realism by playing synthesized audio and performing expressive gestures.

3 Experiment

3.1 Participants

We invited 27 children (15 girls and 12 boys) from the local international school to participate in our workshop. Two sessions were organized, one for each class (13 in the first one and 14 in the second). The participants were 7 to 9 years old ($M = 8.33$, $SD = 0.55$) and came from different parts of the world, all studying English as a second language. However, it is essential to note that most of them are already skilled English speakers. The study followed the ethical and legal guidelines of our university¹.

3.2 User Study Design

To address the research question and explore participants' initial perceptions and interests regarding the

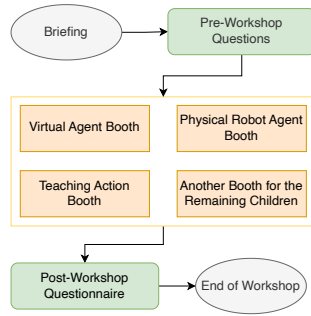
two teachable agents, as well as any relative changes over time, the workshop followed the flow depicted in Figure 3a. It began with an introduction to both agents and explained that the agents were confused about the pronunciation of certain English words and needed the children's help. To identify different agents, we named them after familiar cartoon characters: Jerry for the physical robot agent and Tom for the virtual agent. We also demonstrated how to use the interface to teach the robots. After this initial briefing, participants reported their pre-preference in teaching both agents. The children were then divided into four groups, each comprising 3–4 participants, as organized by their teacher. A within-subject study design was employed, where each group sequentially participated in four booths presented in a pre-defined order. To mitigate the order effect, half of the groups interacted with the Virtual Agent Booth before the Physical Robot Agent Booth, while the other half experienced the reverse order. The four booths included: a Virtual Agent Booth, a Physical Robot Agent Booth, a Teacher Acting Booth (children teaching the pronunciation of their first language to peers within the same group to allow us to observe their natural peer-to-peer teaching behaviour for future study), and a remaining booth with educational tablet activities. The settings for the Physical Robot Agent Booth and the Virtual Agent Booth are shown in Figure 3b. Due to the time limit, the "Post-test" phase of the design in Figure 1 was excluded from the study in both agents conditions. After completing all four booths, participants were invited to answer post-workshop questions to conclude the session.

3.3 Measures

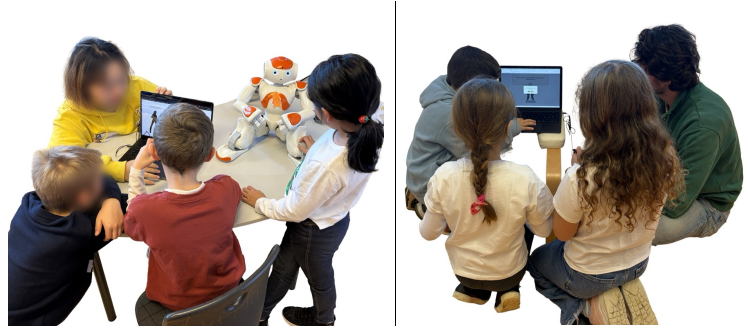
Mispronunciation Recognition Rate. All the children in the study were fluent in English and had a strong ability to recognize mispronunciation. To assess whether children can correctly perceive the mispronunciations generated by the PMP method (RQ1), we added a pop-up window with the question "I think the pronunciation is:" followed by two buttons labeled *Correct* and *Wrong* after each pronunciation by the agent. This feature explicitly encouraged children to evaluate the performance of the teachable agent. This data was logged for the data analysis.

Motivation Score. To evaluate children's motivation to teach the currently designed agents (RQ2), questions adapted from the Intrinsic Motivation Inventory (IMI) [22] were asked after the sessions. It is based on a self-reported 5-point Likert scale, where higher scores indicated stronger learning motivation, and three subscales were chosen for evaluation: *Interest*: Measures intrinsic motivation by assessing enjoyment and engagement during the activity, with the question "I think teaching Tom/Jerry was quite enjoyable"; *Perceived Competence*: Evaluates participants' confidence in successfully completing tasks, with the question "I think I was good at teaching Tom/Jerry"; *Effort*: Reflects the level of effort exerted, correlating with their motivation, with the question "I put much effort into teaching Tom/Jerry". To ensure the questionnaire was easy to understand for children, we replaced

¹This study has received ethical approval from the Human Research Ethics Committee of EPFL under protocol HREC 057-2021.



(a) Workshop Flow



(b) Workshop Setting for Physical Robot Agent (Left) and Virtual Agent (Right)

Figure 3: Workshop Design

the standard numeric Likert scale with a smiley-face scale [10].

Preference Score. To assess the children's preference in teaching the agents (RQ3), the 5-point Likert scale question "I prefer to teach Tom/Jerry" was asked both before and after the workshop.

4 Results

4.1 Performance of the Teachable Agent's Pronunciation Generation System

On Phoneme-level Mispronunciation Projection Method: To assess whether children can correctly perceive the mispronunciations generated by the PMP method (RQ1), as described in Section 3.3, we computed the mispronunciation recognition rate to measure the children's perceived performance of our PMP approach. As shown in Figure 4, the overall recognition rate is 80.6%, with 79.8% for the virtual agent and 81.6% for the physical robot agent respectively, which indicates the proposed PMP method could effectively articulate the mispronunciation, therefore project the pronunciation difficulties to the teachable agent.

On Overall Pronunciation Generation System: Additionally, we computed the F1 score to assess the overall pronunciation generation performance of the teachable agent as Table 1. The F1 scores for the virtual agent ranged from 0.483 to 0.923, while those for the real robots ranged from 0.500 to 0.944. With an overall F1 score of 0.751, in most cases, children were able to correctly distinguish the correct and wrong pronunciation of the agent. But sometimes some pronunciation errors are difficult to recognize; for example in workshop 1 of Group D, star (/stɑ:r/) with an error phoneme /r/ replaced by /æ/, which made the pronunciation /stɑ:æ/.

4.2 Subjective Feedback of Children

4.2.1 Motivation Score. To measure children's motivation to teach the agent in the current design (RQ2), the IMI score of each child was calculated by averaging the applied IMI items as described in Section 3.3. The IMI

score of the virtual agent and the physical agent is presented in Figure 5b. A two-tailed paired Wilcoxon signed-rank test was conducted to compare the motivation of children between teaching the virtual and physical agent. The results revealed a marginally significant difference in children's motivation to teach between two conditions, $W = 8.50$, $z = 1.95$, $p = .050$, with a medium effect size $r = 0.38$. The motivation score for the virtual agent condition is $M = 4.39$, $SD = 0.73$, while the motivation score for the physical agent condition is $M = 4.69$, $SD = 0.49$. We chose the non-parametric statistical test because the data normality was rejected by the Shapiro-Wilk test ($W = 0.75$, $p < .001$).

4.2.2 Preference - Virtual Agent vs. Physical Robot Agent. In addition, to probe the preference of children between two forms of teachable agents, we computed the average preference score of two conditions before and after the experiment as Figure 5a. The Shapiro-Wilk test also rejected the normality of the preference score data ($W = 0.38$, $p < .001$). A Wilcoxon signed-rank test was thus conducted to compare the score between virtual agent and physical robot agent. For the pre-workshop survey, no statistically significant differences exist in the preference score between the virtual agent ($M = 4.72$, $SD = 0.88$) and physical robot agent ($M = 4.95$, $SD = 0.21$), $W = 0.00$, $z = 1.34$, $p = .18$, $r = 0.26$. However, after the children completed the learning by teaching interaction with both forms of agents, a statistically significant higher preference score was revealed by the Wilcoxon test for the physical robotic agent ($M = 4.91$, $SD = 0.29$) compared with the virtual agent ($M = 4.27$, $SD = 1.32$), with medium effect size, $W = 6.00$, $z = 1.99$, $p = .046$, $r = 0.39$.

5 Discussion

5.1 Children's Perception on the Pronunciation Behaviour of the Teachable Agent

The children's ability to identify and respond to the pronunciation behaviours of teachable agents is central to the effectiveness of the learning-by-teaching paradigm.

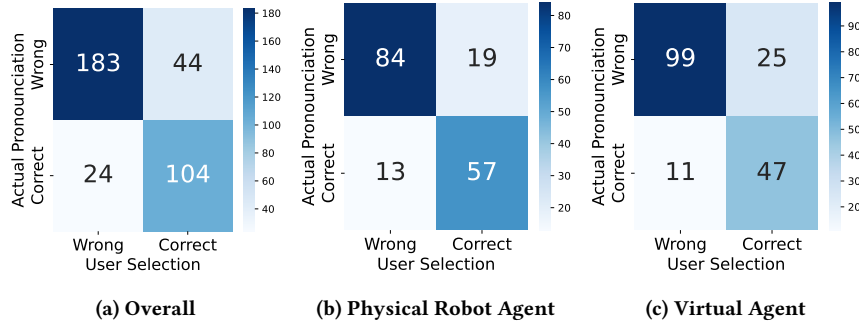
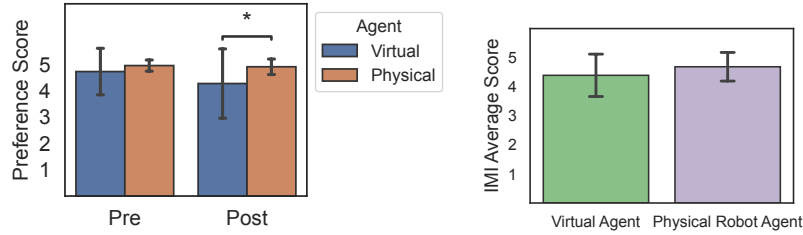


Figure 4: Confusion Matrix of Mispronunciation Recognition

Table 1: Generated Phonetic Mispronunciation Pairs and F1 Scores. V for Virtual Agent, R for Physical Robot Agent. PP for Phoneme Pair and RR for Recognition Rate (Due to the technical issue, the log data in Workshop 1 Group C is incomplete.)

Workshop	Group	V (PP)	V (RR)	V (F1)	R (PP)	R (RR)	R (F1)
Workshop 1	A	/dʒ/ → /d/	0.875	0.778	/dʒ/ → /tʃ/	1.000	0.857
	B	/r/ → /l/	0.714	0.682	/l/ → /oʊ/	0.667	0.719
	C	/l/ → /oʊ/	Nil	Nil	–	Nil	Nil
	D	/l/ → /oʊ/	0.909	0.914	/r/ → /æ/	0.75	0.692
Workshop 2	A	/b/ → /r/	0.286	0.483	/l/ → /oʊ/	0.792	0.786
	B	/l/ → /oʊ/	0.955	0.923	/b/ → /oʊ/	1.000	0.944
	C	/l/ → /oʊ/	1.000	0.889	/b/ → /r/	0.830	0.500
	D	/l/ → /oʊ/	1.000	0.923	/r/ → /ə/	0.800	0.750
Overall	–	–	0.798	0.723	–	0.816	0.777



(a) Preference Score for Different Agent Types in Pre- and Post-Workshop (b) Motivation (IMI) for Different Agent Types in Post-Workshop

Figure 5: Subjective Feedback Result

The study results highlight two key aspects of their perception, which directly influence their engagement and learning outcomes. Firstly, given the fact that all children are fluent English speakers, their reported high overall mispronunciation recognition rate of 80.6% indicates that the PMP method effectively reproduces the phoneme-level mispronunciation on different words, enabling children to identify and address these errors by teaching the agent. Secondly, the high F1 score demonstrates the children's consistent ability to accurately recognize both mispronunciations and correct pronunciations delivered by the designed agent. It suggests the children could perceive the agent's adaptive pronunciation performance of the current prototype, including

improvements or deterioration, throughout the teaching process. Such capability of observable learning progress has been identified as a critical element for engaging learners in the learning by teaching paradigm [28].

5.2 Children's Motivation to Teach

Benefiting from the well-perceived pronunciation behaviour of the teachable agent, we argue that the designed prototype can motivate children to learn in a unique way, beyond the other advantages of the learning-by-teaching method. In the designed activity in Figure 1, since the mispronounced words presented by the agent during the LBT phase differ from those they were tested on in the pre-test phase, children might have less

awareness of their performance and feel more confident. This can therefore provide a more supportive, non-judgmental environment for them to teach. On the other hand, because the phoneme-level mispronunciations are properly projected to the agent, the children can still practice the specific phonemes that are tailored to their needs. Our preliminary experiment findings indicate that the high motivation score aligns with such design choices made. However, due to the absence of a control condition caused by time constraints in our study, it is necessary to further investigate whether the learning-by-teaching setting itself or the specific pronunciation-focused teachable agent design contributed more significantly to enhancing children's motivation.

5.3 Teachable Virtual Agent vs. Teachable Physical Robotic Agent

The comparison between virtual and physical teachable agents revealed interesting differences in their impact on children's motivation. Literature suggests that physical robots often evoke a stronger sense of presence and engagement due to their tangible and interactive nature, fostering richer social interactions compared to virtual agents [14, 26]. This aligns with our findings, where the physical robotic agent received significantly higher preference scores post-interaction compared to the virtual agent. Children also exhibited slightly higher motivation when teaching the robot, possibly due to its realistic gestures and dynamic responses, which reinforced the perception of a "real" student. However, the virtual agent has its own advantages, particularly in accessibility and scalability. It can be deployed without the logistical and financial constraints associated with physical robots. Despite the differences in embodiment, the core pedagogical value of the system lies in its design for both types of teachable agents to support pronunciation learning.

5.4 Design Implications

The results of this work suggest several design implications for developing effective teachable agents for pronunciation learning (RQ4): 1) Accurately analyzing and simulating mispronunciations is key to enabling children to perceive the agent's pronunciation performance and provide meaningful feedback and teaching. As suggested in Section 4.1, subtle errors may be overlooked, especially when the agent's voice lacks sufficient expressiveness. This aligns with prior work emphasizing the importance of exaggerated yet realistic phonetic cues to aid perception and correction [3]. 2) Incorporating dynamic, adaptive pronunciation behaviours that respond to the children's teaching can enhance their sense of agency and investment in the teaching process. 3) Interaction flow and feedback mechanisms are critical. Based on our observation during two workshops, positive reinforcement, such as celebratory gestures, enhanced children's confidence and motivation. Moreover, incorporating a mentor agent for evaluation provided clear guidance, reinforcing learning. 4) Our study findings suggest that

the physical embodiment of the agent may enhance children's motivation, while the selection of the agent's form should also weigh the trade-offs between engagement, accessibility, and cost.

5.5 Future Work and Limitations

This study highlights the potential of teachable agents for pronunciation learning but leaves room for improvement. The current PMP method could benefit from advanced algorithms, including deep-learning-based mispronunciation exaggeration techniques, to enhance error clarity and detection. This refinement would address cases where subtle phonetic distinctions were missed by learners. Additionally, the system's focus on English limits its applicability. Expanding support to other languages, particularly those with complex phonetic structures, would make the approach more versatile and inclusive for multilingual learners. Lastly, while the study examined motivation and engagement, it did not evaluate children's learning gains comprehensively. Future work should include longitudinal studies to measure pronunciation improvements and skill transfer to real-world contexts. Such evaluations would provide deeper insights into the long-term effectiveness and educational impact of teachable agents, ensuring their broader applicability in diverse learning environments.

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